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Paleomagnetism of the Ulkan massif (SE Siberian platform) and the apparent polar wander path for Siberia in late Paleoproterozoic–early Mesoproterozoic times

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ABSTRACT

Paleomagnetic studies on Paleoproterozoic rocks of the Ulkan massif from the south-eastern Siberian platform yielded two ancient paleomagnetic directions and made it possible to calculate poles corresponding to them for two time intervals in the late Paleoproterozoic: (a) ~1732 Ma for volcanogenic-sedimentary rocks of the Elgetey Formation – Plat = 7.1° (−7.1°), Plong = 183.5° (3.5°), d_p = 10.5°, d_m = 16.4° (positive fold, reversals and conglomerates tests); and (b) less than 1720 Ma for granites of the Ulkan Complex – Plat = 42.1° (−42.1°), Plong = 249.4° (69.4°), d_p = 3.4°, d_m = 5.6°. The analysis of the existing paleomagnetic data for Siberia in late Paleoproterozoic–early Mesoproterozoic times allowed a total of 12 new poles to be defined with five of them considered key poles (Table 3), based on which the APWP for Siberia was generated. On the basis of the data presented here, one can conclude that in the context of the Paleoproterozoic supercontinent model, preference is given to the position of Paleoproterozoic north poles of Siberia in the Indian Ocean and Australia, rather than in the Atlantic Ocean.

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1. Introduction

The Siberian (East Siberian) platform of Precambrian age (Fig. 1a), which is almost 4.5 million km^2 in area, occurs at the centre of the Central Asia structure. In the north, east and south it is bounded by the Novaya Zemlya–Taimyr, Verkhoyansk–Kolyma and Mongol–Okhotsk Phanerozoic fold belts, respectively, and in the west by the young West Siberian platform. The structure of the Siberian platform basement is asymmetric. Early Precambrian structures are widespread in its southern part: the Aldan–Stanovoy shield dominates the south-east of the platform; the Akitkan volcano-plutonic belt and the Sharyzhalgai salient are dominant in the south; and the Angar–Kan volcano-plutonic belt and the Yenisei ridge are located in the south-west and west. In the northern part of the platform, these structures occur only within the Anabar massif (shield) and a small part of the Olenek uplift. Meso- and Neoproterozoic sedimentary formations which are divided into two structural stages are widely developed within the Siberian platform along with the Phanerozoic sedimentary cover.

At the end of the last century, a number of authors (Borukaev, 1985; Zonenshain et al., 1990; Tectonics, 1988; Rosen et al., 1994) put forward a model of the Siberian craton formation within the context of the terrane concept, which is based on differences in geological structure of individual blocks of the craton as well as petrological and geochronological data: Angar–Anabar, Tunguska, Olenek, Aldan and Stanovoy (Fig. 1b). According to Zonenshain et al. (1990), Rosen et al. (2006) and Tectonics (2001), prior to ~1.9 Ga there was no single Siberian craton – the above five terranes were separated by the ocean basins. At the end of the Paleoproterozoic they formed two distinct cratons: one encompassing Aldan and Stanovoy terranes, and the second, formed by the collision of the Angar–Anabar, Tunguska and Olenek terranes.

Borukaev (Tectonics, 1988), and later Zonenshain et al. (1990) and Rosen et al. (2006), related the Akitkan and Bilyakchan–Ulkan (Ulkan) volcano-plutonic belts to the major structural sutures that joined terranes of the Siberian craton in the Paleo-Mesoproterozoic (Fig. 1b). The latter are identified by Zonenshain et al. (1990) with continental-margin volcanic arcs. The wide occurrence of such structures in all cratons of the Earth allowed Borukaev (1985) to categorize them as Akitkan-type structures.

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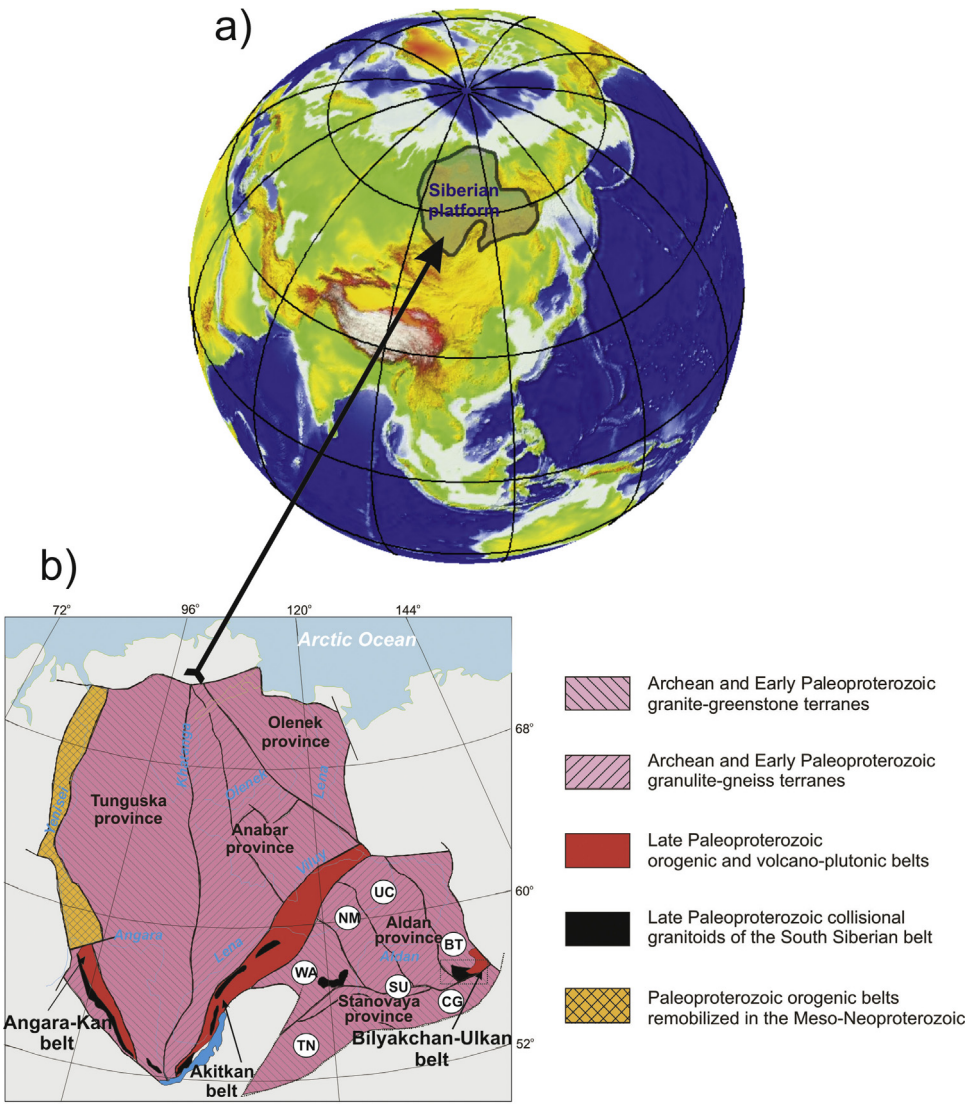


Fig. 1. The Siberian platform in the World (a) and General tectonic scheme of the Siberian craton (b) according to Rosen and Fedorovsky (2001) and Rosen et al. (2006). Terranes: BT – Batomga, CG – Chogar, NM – Nimnyr, SU – Sutam, TN – Tynda, UC – Uchur, and WA – West Aldan.

In his comprehensive review, Khain (2001) considers the Ulkan trough, as well as Bilyakchan and Udokan ones, to be created by rifting. According to him, rifting began ~2.2 Ga ago with the formation of the Udokan depression filled in with a thick series of continental-clastic red beds and granitoids of the Kodar–Kemen Complex and spread eastward (in present-day coordinates) into the Ulkan and Bilyakchan troughs. Originally it was Berezhnaya et al. (1988) who estimated the age of the Chiney sub-Group of the Udokan Group as 2.18 Ga. There is an alternative opinion as to the genesis of the Udokan–Chiney structures – transition of the passive continental margin into an active one (Abramov, 2007).

According to Larin (2008), whose view is conformable with the above but different in the timing of rifting, the Ulkan structure formed "... under the conditions of lithospheric stretching in within-plate environments ... of active rifting due to large-scale upwelling of hot mantle beneath the growing Nuna supercontinents (1.75–1.70 Ga)". Later on this view was developed in Larin (2009) and Larin et al. (2012). There is also a link with the ca. 1750 Ma radiating dyke swarm of Gladkochub et al. (2010) which is focused on the eastern end of the Akitkan belt.

It is shown in Kotov (2003), Rosen and Fedorovsky (2001), Rosen et al. (2006), Smelov and Timofeev (2003) and Tectonics (2001) that the Aldan–Stanovoy shield consists of a number of terranes varying in in-filling (West Aldan, Batomga, Tynda, Nimnyr, Chogar, Sutam, and Uchur; Fig. 1b), separated by the zones of tectonic melange (Kalar, Amga, and Tyrkanda). Complexes that "suture" them, including anorthosites, are developed within these zones and some terranes (Tectonics, 2001). Anorthosites of the Dzhugdzhur massif, which are part of the anorogenic Bilyakchan–Ulkan volcano-plutonic belt, "suture" the Early Precambrian tectonic structures of the south-eastern Siberian platform, and the observed structure of the Aldan–Stanovoy shield took shape in the second half of the Paleoproterozoic (Tectonics, 2001).

To date reliable paleomagnetic data on Tunguska and Anabar provinces of the Siberian craton have been published for the second half of the Paleoproterozoic and Mesoproterozoic, which made it possible to develop the first apparent polar wander path (APWP) for Siberia for this time interval (Didenko et al., 2009a,b; Veselovsky et al., 2009; Vodovozov et al., 2011). Up to now, there have been no reliable paleomagnetic data on this age interval for Aldan and Stanovoy provinces of the Siberian craton.

This paper deals with the following issues: (1) presentation of the first paleomagnetic data on the upper Paleoproterozoic rocks of the Aldan–Stanovoy province; (2) comparison of the defined

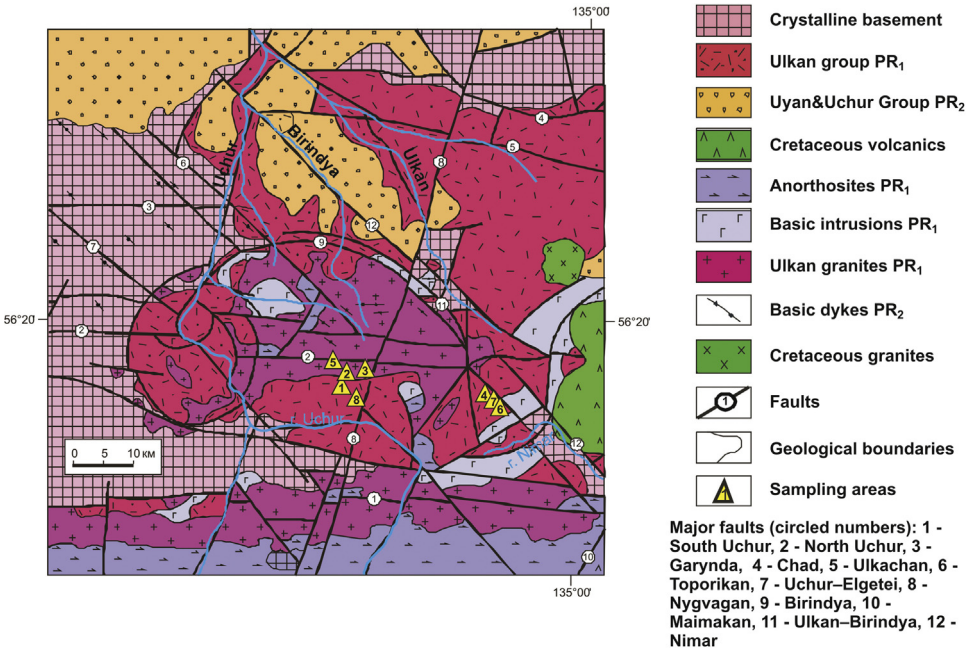


Fig. 2. Geological sketch-map of the Ulkan massif (trough) and the adjacent territory of the Siberian platform.

paleomagnetic poles with paleomagnetic data on Tunguska and Anabar provinces (Didenko et al., 2009a,b; Vodovozov et al., 2011), and (3) presentation of a new model of the APWP for Siberia for the second half of the Paleoproterozoic and early Mesoproterozoic.

2. Brief geologic setting of the study area

Structures of the Ulkan massif occur on the south-eastern edge of the Aldan terrane (Fig. 1b) in the upper reaches of Uchur, Toporikan, Ulkan, and Nimar (Fig. 2). Paleoproterozoic unmetamorphosed rocks here (sedimentary-volcanogenic rocks of the Ulkan Group and granitoids of the Ulkan Complex) are overlain by Mesoproterozoic deposits of the platform cover and rest with angular unconformity on metamorphic rocks of the Aldan–Stanovoy shield and their weathering crust (Fig. 3) thus marking the time of consolidation of the Siberian platform crystalline basement.

Alekseev et al. (1962) were the first to distinguish the Ulkan Group within Toporikan and Elgetey Formations. Later on its composition and volume were repeatedly revised by different authors; in this study V.A. Guryanov's stratigraphic scheme is used (Guryanov, 2007; Didenko et al., 2010; Guryanov et al., 2012), under which the group comprises Toporikan, Ulkachan and Elgetey Formations whose total thickness is over 3 km (Fig. 3). Subvolcanic intrusions and vent extrusions make up a large part of the group (Fig. 2) which together with lava flows are considered as Ulkachan trachybasalt and Elgetey basalt–trachyrhyolite volcanic complexes.

The Elgetey Formation which is over 2 km thick and divided into three sub-formations (Fig. 3) occurs at the top of the Ulkan Group. It is dominated by red trachyrhyolites, trachyrhyodacites and trachydacites. There also occur their tuffs, ignimbrites and lava breccias, pantellerites and comendites; less common are mafic lavas. Sedimentary and tuffaceous-sedimentary rocks occur in lava flows as rare interbeds and layers. The Elgetey Formation, as a rule, rests unconformable either on the Ulkachan or Toporikan Formation, and outside of these occurrences, it overlies the crystalline basement. According to Larin (2008), the age of trachyrhyolites in the Elgetey Formation is assessed at 1720 + 2.5 Ma (U–Pb method on zircons). According to our data (Didenko et al., 2013), the concordant age of zircons (U–Pb) from the Upper Elgetey sub-Formation is 1732 ± 4 Ma (analyses were carried out by E.B. Salnikova and A.B. Kotov in the Laboratory of Isotope Geology at the Institute of Geochemistry and Precambrian Geology, RAS).

Granitoids of the Ulkan alkaline–leucogranite complex make up two large massifs and several small- and mid-size intrusions, which are exposed in different parts of the Ulkan trough and its framing (Fig. 2). The largest two massifs are Ulkan and South Uchur, the former being petrotypical. Ulkan granitoids intruded in three consecutive phases: 1st phase with an age of 1729.7 ± 1.9 Ma (U–Pb zircon) – subalkaline granites and biotite leucogranites, quartz syenites, monzonites, and syenite porphiries; 2nd phase – subalkaline biotite fine-grained leucogranites; 3rd phase with an age of 1724 ± 3.6 Ma (U–Pb zircon) – alkaline granites and quartz aegirine-riebeckite syenites (Didenko et al., 2010). Analytical studies were carried out under the direction of E.V. Bibikova in the Laboratory of Isotope Geochemistry and Geochronology at Vernadsky Institute of Geochemistry and Analytical Chemistry, RAS.

According to Neymark et al. (1992) and (Larin et al., 1997), zircons from first, second and third phase granites of the Ulkan Complex are dated at 1720.8 ± 1.0, 1715.8 ± 2.5 and 1704.6 ± 4.7 Ma (U–Pb zircon), respectively.

Sediments of the lower Mesoproterozoic Uyan Group rest with angular unconformity on rocks of the Elgetey Formation (Fig. 3). The significance of this break is highlighted by the development of the ferruginous–clay–hydromicaceous weathering crust with the Rb–Sr isochron age of 1670 ± 40 Ma (Guryanov, 2007; Kazakov and Knorre, 1973) underlying the Birindya Formation (bottom of the Uyan Group).

To resolve contradictions among the existing models a detailed paleomagnetic study of the Ulkan Group volcanics and the Ulkan massif granites was conducted at eight sites within its central part (Fig. 2; mean coordinates: 56.25° N, 134.67° E).

3. Paleomagnetic studies

3.1. Paleomagnetic research techniques

Field and laboratory paleomagnetic studies and data processing were carried out mainly by standard procedures

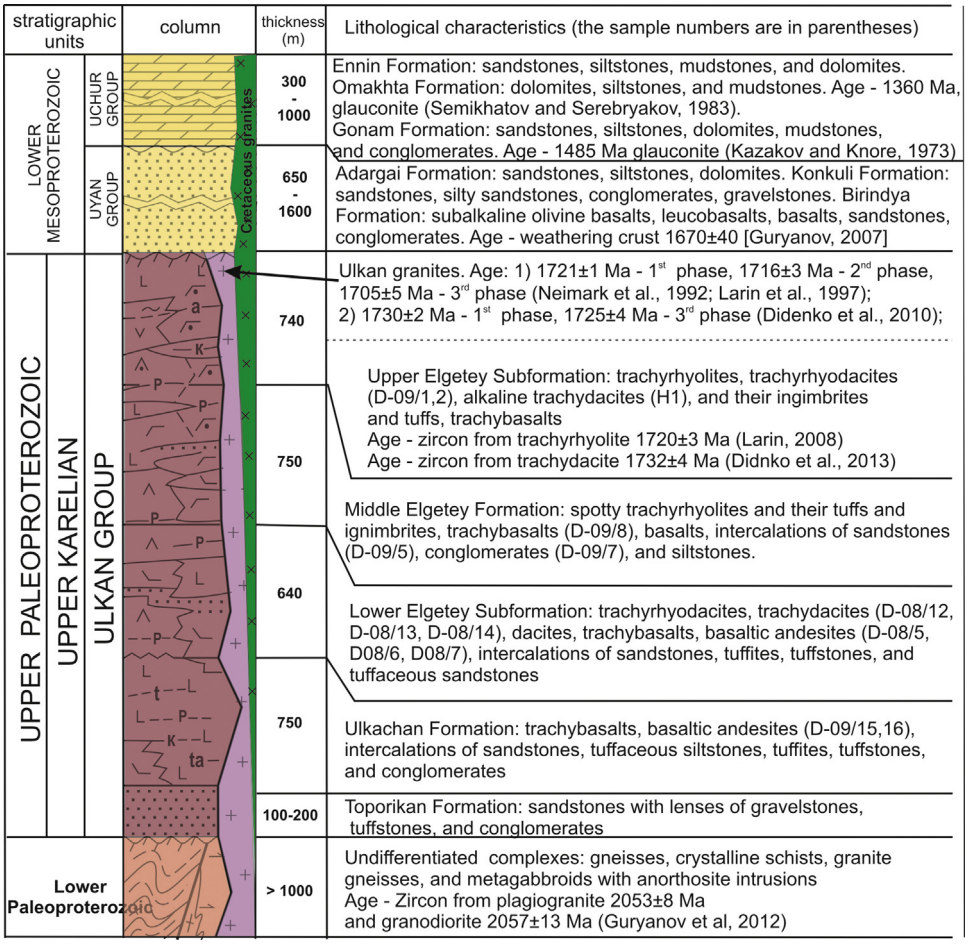


Fig. 3. The Ulkan Group in the Precambrian section of the south-eastern Siberian platform.

(Paleomagnetology, 1982; Pechersky and Didenko, 1995; Butler, 1992; Van der Voo, 1990). Sampling for rock magnetic and paleomagnetic measurements was preceded by measuring the initial magnetic susceptibility (k) directly in outcrops using a Czeck field magnetic susceptibility metre KT-5 (sensitivity of 1×10^{-5} S.I. units). Fine-grained sediments, fine-grained granites, and igneous rocks with k of moderate magnitude were our first choice. During igneous rock sampling the preference was given to quench zones which were recognized both visually and by a regular decline in k values from the centre of bodies outwards. In laboratory conditions, measurements of k for all the samples were repeated using a Czech magnetic susceptibility metre MFK-1FA (sensitivity of 5×10^{-8} S.I. units).

A magnetic compass was used for sample and bedding element orientation and magnetic declination was taken into account when determining mean paleomagnetic directions. Rock samples were cut into 2 cm cubes in the laboratory using a diamond saw.

To estimate the age of the natural remanent magnetization (NRM) in rock samples we have succeeded in applying the intraformational conglomerate test, unfortunately, in a strongly reduced form, for rocks of the lower sub-formation of the Elgetey Formation as well as fold and reversal tests. We failed to apply any direct field tests for granitoids of the Ulkan Complex.

Magnetic parameters which characterize the structure, concentration, and carriers of magnetism were examined using a number of representative samples of the studied rocks. Initial and repeated temperature dependences of saturation magnetization (Ms) were primarily studied to determine the Curie point (Tc) and the stability of magnetization carriers to heating. All the samples were subjected to the complete stepwise thermal demagnetization up to 570–670 °C in the furnace with the four-layer permalloy shielding (the magnetic field in the furnace was less than 15–20 nT). Each time samples were heated their position in the furnace was changed randomly to decrease the impact of laboratory magnetizing. A heating step depended on the behaviour of magnetization in the study and varied progressively from 50 to 100 °C at low temperatures to 20–30 °C at medium temperatures and to 5–10 °C at high temperatures. Usually the number of thermal cleaning steps ranged from 15 to 20. NRM studies were done with a Czech spinner magnetometer JR-6a (bulk magnetic moment sensitivity is 2.4×10^{-6} A/m) mounted in the Helmholtz coils. Samples were transferred from the furnace to the measuring instrument in the μ-metal container.

The data set obtained by demagnetization was analyzed using the PMGSC software v.4.2 (Enkin, 2004). This software comprises standard tools for processing, displaying and analysing palaeomagnetic data, including principal component analysis (PCA; Kirschvink, 1980) and remagnetization circle analysis (RCA; McFadden and McElhinny, 1988). The fold test was performed using the PMGSC software (Enkin, 2004) and reversal tests (McFadden and McElhinny, 1990). T. Torsvik's GMAP software was used (Torsvik and Smethurst, 1999) to trace tectonic block movements on the sphere and to generate apparent polar wander paths (APWPs).

3.2. Magnetization carriers of the studied rocks

The *Ms-T* dependence of the studied, more magnetic rocks, such as samples of trachydacite of the Elgetey Formation (Fig. 4a) and

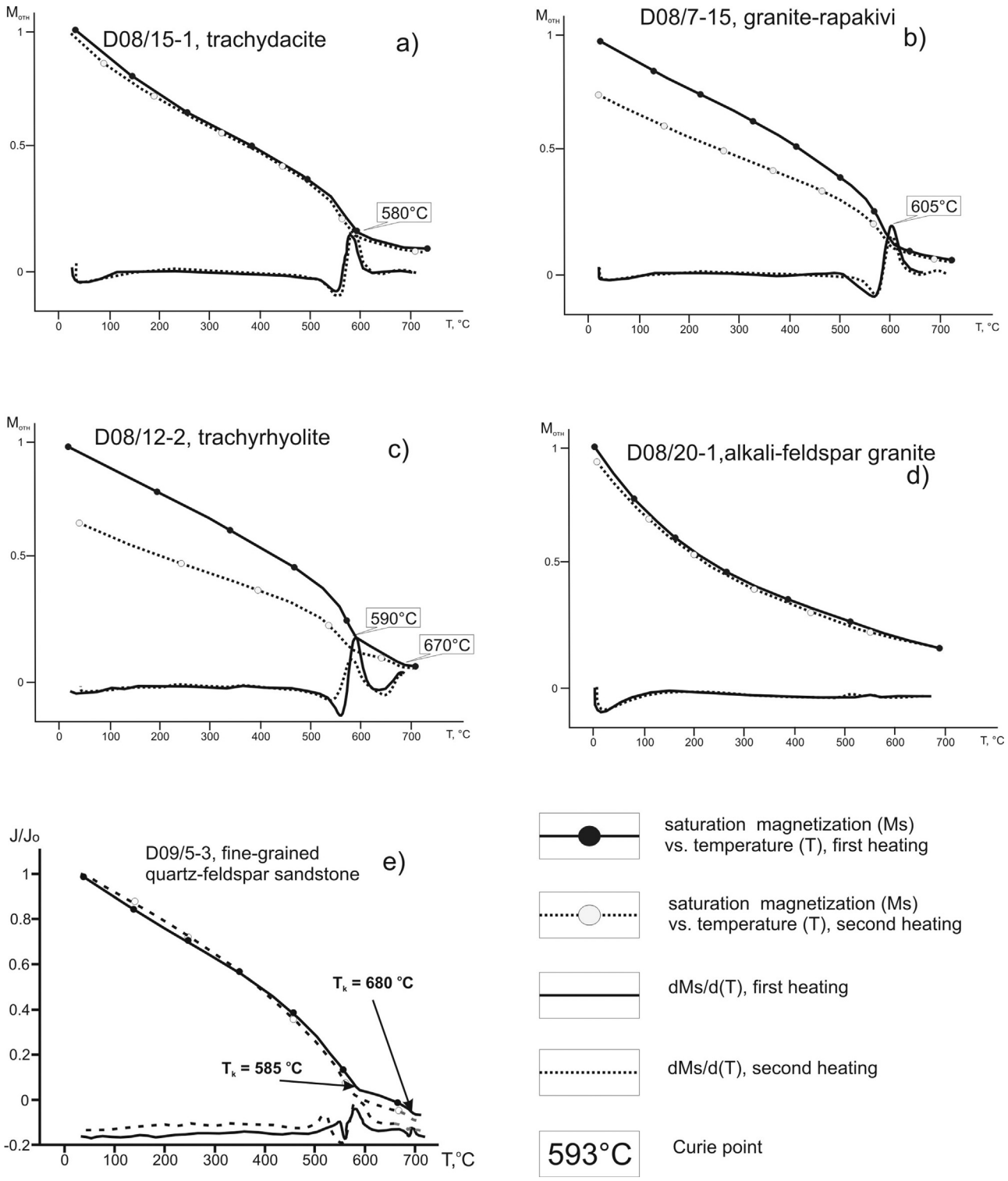


Fig. 4. Magnetic mineralogy. Saturation magnetization vs. temperature (a–e).

rapakivi-granite (Fig. 4b), takes a single-phase form with Curie points between 578 °C and 605 °C. Curie points are well defined on the second derivative curves of *Ms-T* dependences of these samples – a typical "kink" in the above temperature range. These samples exhibit a high stability to the laboratory heating to 700 °C. First, *Ms* values before and after heating are almost equal; second, Curie points when reheated have the same values (Fig. 4a). Behaviour of rapakivi samples when heated in the laboratory is somewhat different: there is a small (10–25%) decrease in *Ms* values after heating, and when reheated, along with the first magnetic phase identified at the first heating, there is a second magnetic phase with haematite temperatures (Fig. 4b). A decrease in *Ms* after heating and the occurrence of hematite at repeated heating testify that some cation-deficient magnetite ($Tc \approx 605$ °C) changes to hematite ($Tc \approx 675$ °C).

Among volcanics of the Elgetey Formation there occur specimens with two-phase magnetic structure, which is inherent, mainly, in acid varieties (Fig. 4c). The first magnetic phase of the

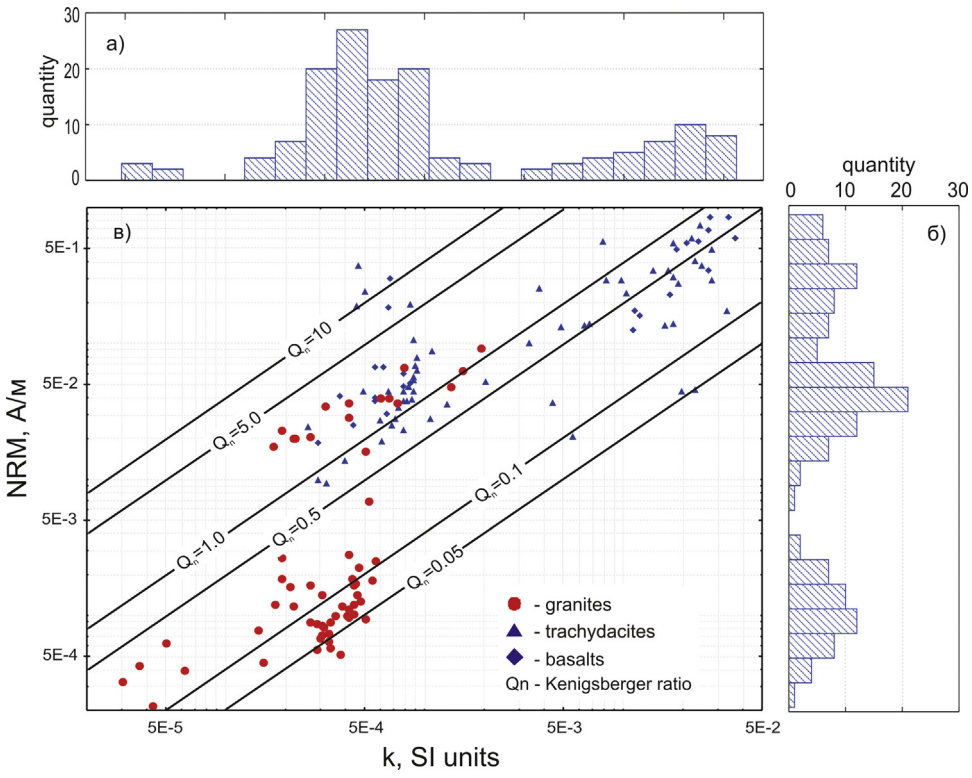


Fig. 5. Natural remanence magnetization (NRM) vs. magnetic susceptibility (k), and its distribution.

trachyrhyolite sample corresponds to the cation-deficient magnetite with $Tc \sim 590\,^{\circ}C$ and the second – to hematite ($Tc \sim 670\,^{\circ}C$). Both of these magnetic phases are clearly seen on *Ms-T* dependences of the first and second heating. *Ms* of such samples decreases appreciably after the first heating to 50–70% of the initial value (Fig. 4c). We believe this is related to the transition of a portion of the cation-deficient magnetite to hematite during laboratory heating.

The third type of the igneous rocks studied is their weakly magnetic varieties with an almost hyperbolic appearance of the *Ms-T* dependence. As an example we refer to Fig. 4d which presents the *Ms-T* dependence of the 2nd phase alkaline feldspathic granite of the Ulkan Complex. The figure shows that following the first and the second heating a small magnetic phase with *Tc* close to magnetite (a gentle kink on the second derivative curves of *Ms-T* relationship for the sample) stands out against the background of the paramagnetic phase. Samples with extremely low concentration of ferromagnetic minerals (a few hundredths of a percent) are characterized by paramagnetic-type curves.

The *Ms-T* dependence of the studied sedimentary rocks of the Elgetey Formation has practically the same shape. As an example we referred to the *Ms-T* dependence for the fine-grained quartz-feldspar sandstone (Fig. 4e), wherein magnetic minerals are presented by two phases: the first either by magnetite or cation-deficient magnetite and the second by hematite. Both of these magnetic phases are recorded on *Ms-T* dependences of the first and the second heating. *Ms* of sediment samples following the first heating remains essentially unchanged (Fig. 4d), suggesting the stability of the Elgetey sediments to laboratory heating.

Values of initial magnetic susceptibility (k) and natural remanent magnetization (*NRM*) of the studied specimens vary within a wide range (Fig. 5) – more than three orders of magnitude: k from 3×10^{-5} to 5×10^{-2} S.I. units, *NRM* from 2×10^{-4} to 1 A/m. The distribution of these values is non-monotonic and non-unimodal, analysis of the *NRM-k* chart and their distributions suggests three groups of samples (Fig. 5): strongly magnetic ($NRM > 8 \times 10^{-2}$ A/m, $k > 3 \times 10^{-3}$ S.I. units), moderately magnetic ($5 \times 10^{-3} < NRM < 8 \times 10^{-2}$ A/m, $1 \times 10^{-4} < k < 3 \times 10^{-3}$ S.I. units) and weakly magnetic ($NRM < 5 \times 10^{-3}$ A/m, $k < 1 \times 10^{-4}$ S.I. units). We believe that such a significant variation of k is, first of all, due to differences in concentrations of ferrimagnetic minerals in these samples, whereas the value of *NRM* depends not only on the concentration of magnetic minerals, but also on their structural features – the linear correlation coefficient of these two variables is 0.74. The group of strongly magnetic samples comprises mafic volcanics of the Elgetey Formation. The moderately magnetic mineral group comprises all the studied felsic and a portion of mafic volcanics of the Elgetey Formation as well as rapakivi-type granites. And the weakly magnetic mineral group comprises alkaline feldspathic granites and sediments of the Elgetey Formation.

The Königsberger ratio (Q_n) of the studied samples varies within a wide range: from 0.05 to 20 for the felsic and a portion of mafic volcanics of the Elgetey Formation, and from 0.035 to 4 for the rapakivi-type granites (Fig. 5). A significant proportion of granites and volcanics of the Elgetey Formation have $Q_n > 1$, suggesting the predominance of the remanent magnetization over the induced magnetization in these rocks, on the one hand, and their magnetic rigidity, on the other. The latter fact is favourable for preserving the ancient remanent magnetization.

3.3. Paleomagnetic characteristics of granitoids

Results of thermal demagnetization up to 600–620 °C of 120 granitoid samples of the Ulkan Complex are given in Table 1 and Fig. 6. The *NRM* behaviour of 27 samples during their stepwise thermal cleaning is unstable and will not be analyzed further. This behaviour is characteristic of extremely magnetically soft specimens with $Q_n < 0.1$ (Fig. 5). The *NRM* behaviour of the remaining 93 samples during thermal magnetic cleaning is interpretable. At least 2–3 paleomagnetic components are normally isolated from

Table 1
Characteristic remanence directions of the Ulkan Complex granites.

Site	Location		Rock type	*N*	In situ		*k*	α_{95}
	Lat (N)	Long (N)			Dec	Inc		
D08/7	56.24	134.42	Rapakivi-granites	17/23	32.2	55.2	12.3	10.6
D08/8	56.29	134.50	Alkali granites, medium-grained	9/12	43.0	23.5	7.5	20.1
D08/18	56.29	134.49	Alkali granites, medium- and fine-grained	14/17	41.6	41.1	18.1	9.6
D08/19	56.30	134.48	Alkali granites, medium- and fine-grained	21/30	44.4	37.2	16.8	8.0
D08/20	56.30	134.47	Alkali granites, medium- and fine-grained	23/24	58.7	36.3	13.4	8.6
D08/22	56.32	134.38	Granites, medium-grained	9/14	64.1	27.4	9.2	17.9
	56.29	134.46	Mean by sites	6/6	48.3	37.2	31.6	12.1
	56.29	134.46	Mean by samples	93/120	46.8	39.8	10.5	4.7

N indicates the ratio of samples used in the statistical analysis versus studied; Dec and Inc are the declination and inclination of the site-mean direction (in°); *k* is Fisher (1953) concentration parameter; α_{95} is the radius of the cone of 95% confidence about the mean direction (in°).

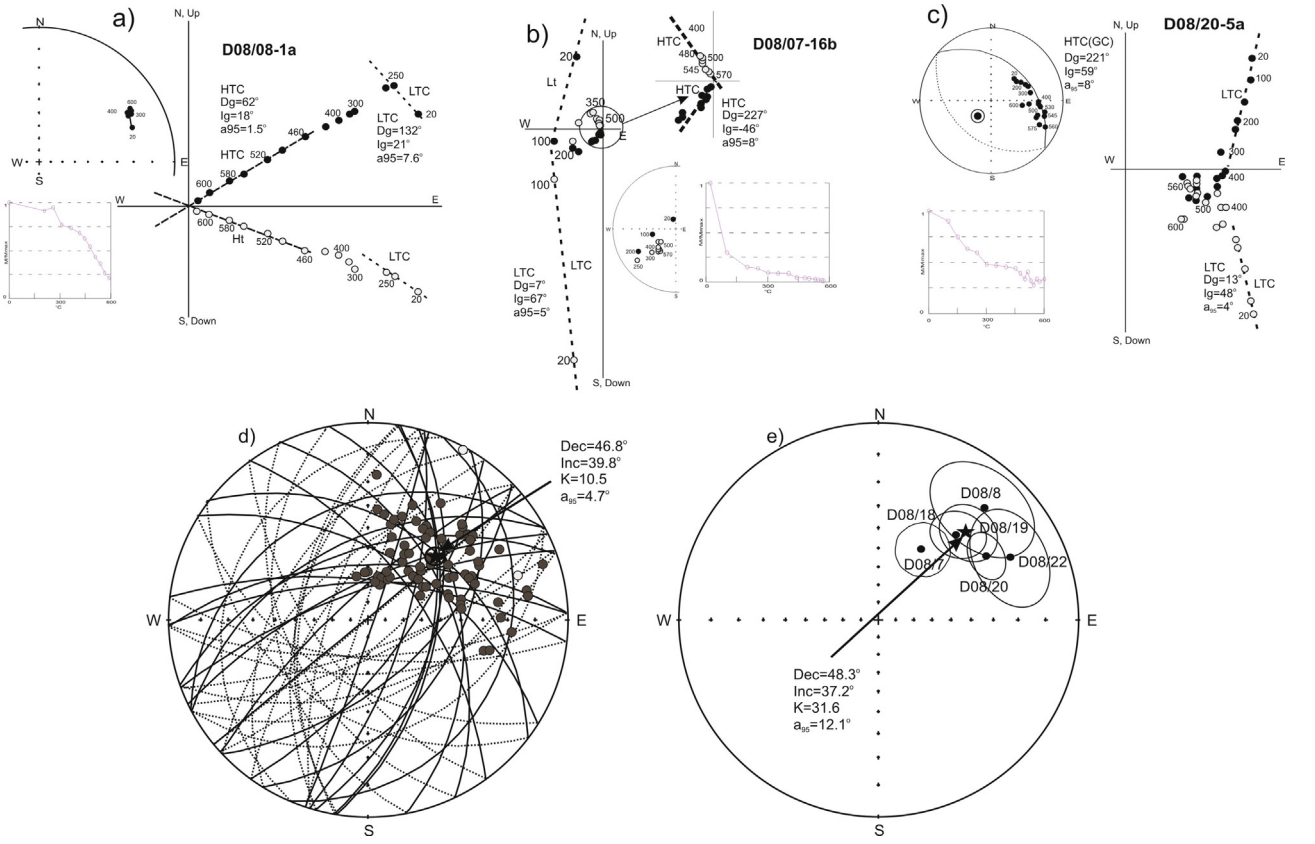


Fig. 6. Examples of NRM stepwise thermal demagnetization (orthogonal vector projections (Zijderveld, 1967), stereogram and intensity curves) for granite samples (a–c) and stereographic projection for sample directions (d), and site mean directions (e) with the cone of 95% confidence about the mean direction for granitoids of the Ulkan massif. Solid (open) symbols are: (1) horizontal (vertical) plane on orthogonal projections; (2) upper (lower) hemisphere on stereogram. Star denotes the mean direction.

each sample. The direction of a high-temperature component (HTC) was resolved for 69 of the 93 samples by principal component analysis (PCA) and for 24 samples by circles of remagnetization. As an example we can refer to the results of thermal demagnetization of three granite samples (Fig. 6) on the orthogonal Zijderveld diagrams (Zijderveld, 1967).

An almost perfect Zijederveld plot was constructed for sample D08/08-1a (Fig. 6a). Here two components are isolated, the direction of the low-temperature component (LTC), isolated by three points from 20 to 250 °C (Dec = 132°, Inc = 21°), being different from the direction of the present-day geomagnetic field in the study area. The *NRM* high-temperature component of the sample (Dec = 62°, Inc = 18°) is isolated by seven points between 460 °C and 610 °C and passes through the origin.

Sample D08/07-16b (Fig. 6b) exhibits a more complex behaviour of *NRM* during thermal magnetic cleaning: (a) the LTC (Dec = 7°, Inc = 67°) being in rough agreement with the direction of the present-day geomagnetic field, is isolated between 20 °C and 200 °C; (b) a smooth inflection is observed then to temperatures 350 °C–400 °C; and (c) the HTC is separated by six points between 420 °C and 570 °C and its direction is Dec = 227°, Inc = −46°. When directions of HTCs of this and the previous sample are compared, it is evident that they are different in polarity. There are but three samples for which such a component is isolated (third quadrant and negative inclination), whereas the number of samples with the opposite direction of the HTC (first quadrant and positive inclination) is significantly higher, that is, 66.

Among granitoid samples there is also a third group, for which the HTC can be isolated only by the circles of remagnetization. An example shows the *NRM* behaviour of sample D08/20-5a during thermal magnetic cleaning (Fig. 6c): (a) the LTC (Dec = 13°, Inc = 48°) being very close in its direction to the direction of the

present-day geomagnetic field, is isolated between 20 and 300 °C; (b) then, against the background of some scattering on the great circle arc there is a regular arrangement of vectors during cleaning to 560 °C (direction of the normal to the plane of the great circle is Dec = 221°, Inc = +59°). A total of 24 granitoid samples exhibit such behaviour.

Table 1 gives site-mean directions of *NRM* high-temperature components of granitoids of the Ulkan Complex as well as their distribution on the stereogram (Fig. 6d and e). Directions of the HTC both by samples (Fig. 6d) and their site-mean (Fig. 6e) are well grouped around the general mean with Dec = 47–48°, Inc = 38–39°; the results of various options for calculating the mean direction of the component are being essentially the same (Table 1). We define this direction as characteristic for granites of the Ulkan Complex. The paleomagnetic pole calculated by the mean direction for 93 samples and used by us in subsequent constructions has the coordinates: Plat = −42.1° (+42.1°), Plong = 69.4° (249.4°), d_p = 3.4°, d_m = 5.6°.

3.4. Paleomagnetic characteristics of rocks of the Elgetey Formation

More than 260 oriented rock samples of the Elgetey Formation have undergone thermal demagnetization, the results of which are presented in Table 2 and Figs. 7 and 8. To substantiate the paleomagnetic reliability of the isolated characteristic magnetization for the rocks of the formation only one test has been carried out to full extent, that is, a fold test, while the other two types, conglomerate and reversal tests, were carried out in a somewhat reduced form.

3.4.1. Conglomerate test

The intraformational conglomerate test was performed on a small number of samples. This is due, firstly, to the small size of pebbles in the studied exposure of conglomerates, and, secondly, to their strongly weathered condition which resulted in their crumbling when cut. Only 6 pebble samples were fully demagnetized thermally. Fig. 7a gives a Zijderveld plot for sample D-09/07-6a; two components are isolated here. The direction of the *NRM* low-temperature component is Dec = 26°, Inc = 64° in geographic coordinates, which is close to the direction of the present-day geomagnetic field in the study area; that of the HTC is Dec = 16°, Inc = −14° and it is isolated between 460 °C and 595 °C by seven points and passes through the origin.

The test of pebbles from the intraformational conglomerates yielded the following results (Table 2): (1) the LTC mean direction is Dec = 346°, Inc = 56° in geographic coordinates and the normalized resultant vector is 0.872; (2) the HTC mean direction is Dec = 10°, Inc = −27° with the normalized resultant vector of 0.605; and (3) the critical value of the normalized resultant vector for a set of six samples is 0.642 (Shipunov, 2000). Thus, the conglomerates test for the *NRM* low-temperature component can be thought of as negative and that for the high-temperature component as positive. Notice that six samples are not enough for a full test.

3.4.2. Thermomagnetic cleaning oriented samples of the Elgetey Formation

The *NRM* behaviour of sediments and volcanics of the Elgetey Formation during thermal cleaning is different. Samples are combined into four groups as a matter of convenience in presenting the data.

1. Samples with a single-component *NRM*, the component being observed in two options of polarity: (a) the fine-grained sandstone sample from the bottom of the upper sub-formation of the Elgetey Formation – Dec = 161°, Inc = 26° (Fig. 7b); (b) the trachyrhyolite sample from the flows overlying the sediments – Dec = 324°, Inc = −42° (Fig. 7c); and (c) the trachydacite sample from the lower sub-formation of the Elgetey Formation – Dec = 130°, Inc = 38° (Fig. 7d). In the latter case, the *NRM* behaviour is more "noisy", the accuracy of resolving the component direction is 8°, whereas in the first two cases – 2°.
2. Samples whose *NRM* is a sum of two and more components: (a) the trachydacite sample from the lower sub-formation of the Elgetey Formation with LTC (Dec = 35°, Inc = +36°) and HTC (Dec = 316°, Inc = −27°) (Fig. 7e); and (b) the trachydacite sample from the same part of the section with LTC (Dec = 36°, Inc = +71°), MTC (Dec = 239°, Inc = −59°) and HTC (Dec = 327°, Inc = −17°) (Fig. 7f).
3. A large group of samples whose *NRM* high-temperature component is isolated by the circles of remagnetization. As an example we refer to the Zijderveld plot and the stereogram with the distribution of vectors during thermal cleaning of sample D08/15-3-1b (Fig. 7g). The LTC with Dec = 128°, Inc = +70° is clearly defined on the Zijderveld plot, while the HTC does not "migrate" towards the origin. The vectors of the stereogram characterizing the latter lie on the arc of the great circle the normal to the plane of which is Dec = 238°, Inc = 19°.
4. The last group comprises samples whose *NRM* has a significant proportion of the component (until complete demagnetization) of a similar direction as the characteristic magnetization of granitoids of the Ulkan Complex (see below). As an example, Fig. 7h presents a diagram of the sample from the basalt flow of the lower sub-formation of the Elgetey Formation whose HTC direction is Dec = 26°, Inc = 26°. Samples of this group with the *NRM* direction close to the "granite direction" were excluded from further analysis.

3.4.3. Fold test

Table 2 and Fig. 8 present site-mean directions of the *NRM* high-temperature component of the first three groups of rocks of the Elgetey Formation which yielded the "non-granite" direction. Analysis of distributions in stratigraphic (Fig. 8b) and geographic (Fig. 8a) coordinates of individual HTC directions at the sample level and of mean directions at the site-mean level showed a significant increase in grouping in the stratigraphic system. In the case of statistics at the sample level, Ks/Kg = 2.14, at the site-mean level, Ks/Kg = 2.53 (Table 2). The best grouping is achieved at 108 ± 12% stepwise unfolding (Fig. 8c) suggesting the predominance of the pre-folding "pre-granite" component in the *NRM* of the rocks of the Elgetey Formation.

3.4.4. Reversal test

The "pre-granite" age of the isolated HTC is supported by the fact that the HTC direction of samples from site D08/12 (Table 2) differs by almost 180° from the mean directions of other five sites. Even though the observed value ($\Delta_o b_s$ = 7.8°) and the value critical for this set of samples ($\Delta_{cri}t$ = 39.9°) differ essentially between directions of the two polarities due to the insufficient number of samples with the reverse component, the reversals test is characterized as indeterminate (McFadden and McElhinny, 1990). For purposes of further interpretation, the HTC mean direction of all the samples in the stratigraphic system is used as characteristic ancient magnetization of rocks of the Elgetey Formation: Dec = 115°, Inc = 42° (Table 2). The paleomagnetic pole calculated based on this direction has the coordinates: Plat = −7.4° (+7.4°), Plong = 10.5° (195.5°), d_p = 3.8°, d_m = 6.2°.

4. Age of paleomagnetic directions of the Ulkan massif

An investigation into rock magnetic properties revealed that: (1) cation-defficient magnetite with Curie points between 580 and

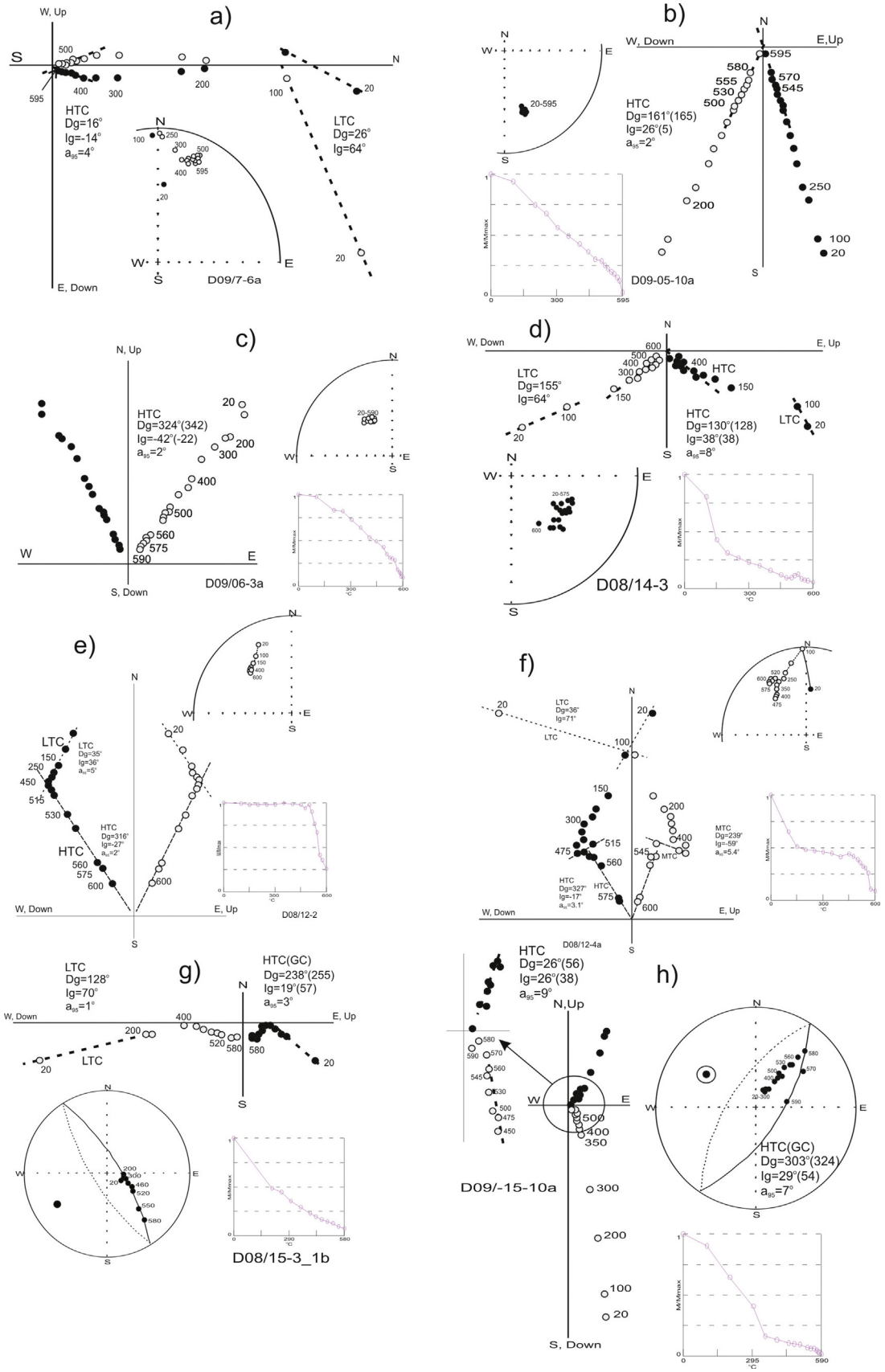


Fig. 7. Examples of NRM stepwise thermal demagnetization (orthogonal vector projections (Zijderveld, 1967), stereogram and intensity curves) for the Elgetey Formation samples. Solid (open) symbols are: (1) horizontal (vertical) plane on orthogonal projections; (2) upper (lower) hemisphere on stereogram.

Table 2
Remanence directions of the Elgetey Formation rocks.

Site	Location		Rock type	*N*	Bedding	In situ		*k*	a_{95}	Tilt-corrected	
	Lat (N)	Long (E)				Dec	Inc			Dec	Inc
D09/7	56.22	134.71	Conglomerate (LTC)	6/17	198/32	346.3	56.4	6.5	22.4	279.8	72.9
D09/7	56.22	134.71	Conglomerate (HTC)	6/17	198/32	10.3	26.9	2.1	62.3	4.8	58.4
*D-08/1	56.24	134.42	Dacite, trachydacite	21/38	165/33	42.2	65.1	7.9	12.0	105.2	57.5
*D-08/3	56.24	134.42	Porphyritic dacite	21/35	165/32	23.9	70.4	93.9	3.3	114.0	65.1
D-08/5	56.25	134.42	Diabase	20/24	165/33	195.4	37.3	30.0	6.1	186.0	11.4
*D-08/6	56.25	134.42	Porphyritic dacite, andesite	7/25	165/32	177.8	56.0	11.5	18.6	167.8	25.3
*D-08/12	56.24	134.67	Trachydacite, trachyrhyolite	12/20	20/23	310.5	−32.9	102.8	4.3	294.9	−42.4
*D-08/14	56.25	134.67	Trachydacite, trachyrhyolite	11/12	33/2	115.4	43.4	28.5	8.7	113.0	43.6
*D-08/15	56.26	134.67	Trachydacite, trachyrhyolite	8/10	38/43	148.3	21.9	46.6	8.2	124.2	37.8
D-09/1	56.18	134.71	Porphyritic trachydacite	12/25	175/37	234.8	−0.8	14.6	11.8	238.8	−11.3
D-09/3	56.21	134.69	Porphyritic trachydacite	7/28	190/35	61.6	12.6	24.6	12.4	73.2	25.0
D-09/5	56.22	134.71	Fine- and medium-grained sandstone	12/15	205/32	143.4	15.9	12.5	12.8	145.5	−6.5
D-09/6	56.22	134.71	Trachydacite, trachyrhyolite	8/12	205/33	159.5	38.1	20.9	12.4	166.8	9.1
D-09/15	56.22	134.41	Porphyritic basalt	12/16	165/33	33.5	−6.6	9.5	14.9	34.2	9.3
	56.23	134.59	Mean by all sites	12/12	k_{max} = 119 ± 19%	111.4	47.2	2.77 3.28	32.3 28.5	119.4	36.8
	56.23	134.59	Mean by samples	151/260		109.0	55.0	2.71 3.15	6.9 6.4	121.6	41.1
	56.25	134.55	*Mean by selected sites	6/12	k_{max} = 110 ± 12%	125.5	57.5	5.42 20.20	31.7 12.8	122.4	46.4
	56.25	134.55	*Mean by selected samples	80/260		105.6	64.7	5.13 10.98	7.8 5.0	115.4	42.4

LTC and HTC are low-temperature and high-temperature components, accordingly; sign "*" denotes sites and average directions on them, used for calculating the characteristic remanence direction. For other designations see Table 1.

600 °C is the main carrier of magnetization among granitoids of the Ulkan Complex; (2) magnetite, cation-defficient magnetite with Curie points between 590 and 610 °C and occasionally hematite are the main carriers of magnetization among volcanics of the Elgetey Formation; (3) studied samples were characterized by stable behaviour (without newly formed magnetic minerals) when heated to 600–700 °C; and (4) a major part of samples of granitoids and volcanics alike have $Q_n > 1$, that is, they are magnetically rigid enough to maintain magnetization with time.

The main criteria for the quality of paleomagnetic data used to solve tectonics problems are (Pechersky and Didenko, 1995; Van der Voo, 1990): (1) a sufficient number of independently oriented samples and satisfactory statistical distribution parameters of interpretable magnetization vectors of the sample collection; (2) full and detailed demagnetization accompanied by PCA; (3) positive tests of paleomagnetic stability; (4) a well-defined age of rocks and correspondence of one of the *NRM* components to this age; (5) structural control of the studied rocks and their tectonic similarity (coherence) to the craton or other tectonic block under study; and (6) no evidence of regional remagnetization. The information on the first three criteria for the resolved ancient paleomagnetic directions is given in the previous section, the last three criteria as applied to the granites of the Ulkan Complex and sedimentary-volcanogenic rocks of the Elgetey Formation are discussed below.

No special research on the thermal history of the Ulkan Complex granites has been carried out but judging by the cooling rates of granitoid intrusions of the Shumikha Complex (Didenko et al., 2005), which have a similar size and are classified as A-type as well as by other estimates of cooling rates for granite intrusions, granites had to pass the threshhold of 578 °C (the Curie point for magnetite) within 5–10 Myr after the closure of the U–Pb (−750 °C) isotopic system in zircons. The literature suggests that the cooling rate of granite intrusions in this temperature range (750–300 °C) can be 11°/Ma (Shen et al., 2000) and from 30°/Ma early in the cooling process to 10°/Ma at the close of the cooling (Pitcher, 1997).

According to the previously published geochronological data on granites of the Ulkan Complex (Neymark et al., 1992; Larin et al., 1997), the age of zircons from first-, second- and third-phase granites of the Ulkan Complex is 1720.8 ± 1.0, 1715.8 ± 2.5 and 1704.6 ± 4.7 Ma, respectively. The age of the Ulkan Complex granites from our data (Didenko et al., 2010) is 1729.7 ± 1.9 Ma (1st phase) and 1724.7 ± 3.6 Ma (3rd phase). We assume that the age of the *NRM* high-temperature component of the Ulkan Complex granitoids should be less than 1720 Ma at an average rate of their cooling of about 20°/Ma.

The second important issue for the subsequent interpretation of the paleomagnetic direction for granitoids, and which requires a comment, is the assumption of the hypothesis of no significant rotation of the Ulkan massif about the horizontal axis after the intrusion of granitoids. Sediments of the lower Proterozoic Uyan Group rest on rocks of the Elgetey Formation, granitoids of the Ulkan Complex and their weathering crust with Rb-Sr age of 1670 ± 40 Ma (Guryanov, 2007) (Fig. 3). Ulkan deposits accumulated in the inland basin in the fluvial-deltaic, littoral and sublittoral sedimentary

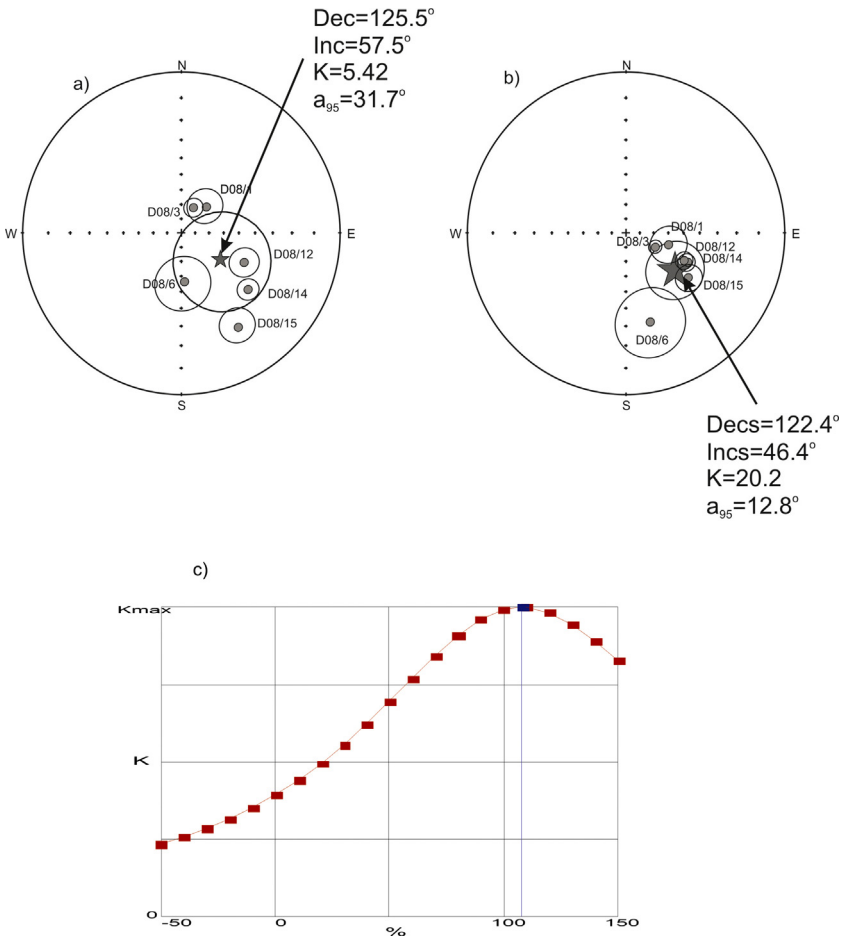


Fig. 8. Stereographic projection in situ (a) and tilt-corrected (b) for selected site mean directions for the Elgetey Formation, and statistical results from stepwise unfolding (c; calculated after Enkin, 2004). Solid (open) symbols are upper (lower) hemisphere on stereogram. Star denotes the mean direction.

environments (Khudoley et al., 2001) are characterized by gentle, almost horizontal bedding (Karsakov et al., 2002; Semikhatov and Serebrjakov, 1983). Thus, there are reasons to assume that granitoids had no rotations about the horizontal axis after 1670 Ma.

The concordant age of zircons from the Upper Elgetey sub-Formation (Didenko et al., 2013) was previously estimated as 1732 ± 4 Ma. Based on the above data on the HTC paleomagnetic stability of rocks in the formation, its age can be characterized as pre-folding and "pre-granite", i.e. ~1732 Ma, which is close to the age of rocks from the upper sub-formation of the Elgetey Formation.

5. A new APW path for Siberia at the end of the Paleoproterozoic–beginning of the Mesoproterozoic

The first version of the APWP for Siberia at the end of the Paleoproterozoic (Didenko et al., 2009a,b) was developed solely on the basis of our own new paleomagnetic data on Akitkan and North Baikal ridges, Sharyzhalgai and Angar–Kan salients of the Siberian platform. Then (Veselovsky et al., 2009; Vodovozov, 2010; Vodovozov et al., 2011) came up with new versions of the APWP for Siberia at the end of the Paleoproterozoic–Mesoproterozoic involving data on the Anabar Massif, Olenek uplift, and Aldan–Stanovoy shield. And new data on the Ulkan massif (SE of the Aldan–Stanovoy shield) prompted us to turn again to the issue of defining the APWP for Siberia at the end of the Paleoproterozoic–beginning of the Mesoproterozoic.

5.1. Analysis of the available data and generation of the APWP

A total of 31 independent palemagnetic measurements for Siberia were obtained while reviewing GPDB-4.6 (Pisarevsky, 2005) and newly published results (Veselovskiy, 2006; Veselovsky et al., 2009) for the above indicated time interval (Table 3). Some of the measurements were made on rocks of the same formations but, except the Kotuikan Formation of the Anabar massif, their paleomagnetic directions differ essentially: (1) 2A for the Gonam Formation (Veselovsky et al., 2009) and 2B for the Konkuli and Gonam formations (Pisarevsky, 2005) of the Uchur–Maya sedimentary basin of the Aldan–Stanovoy province; (2) 3A (Veselovsky et al., 2009) and 3B (Pisarevsky, 2005) for the Burdur Formation of the Anabar massif; and (3) 4A (Veselovsky et al., 2009) and 4B (Pisarevsky, 2005) for the Kotuikan Formation of the Anabar massif. In Fig. 9a is shown the geographical position of the paleomagnetic targets.

Reliability analysis of the paleomagnetic results is not favourable for several reasons, the main being the accuracy of determining the age of the paleomagnetic direction interpreted. According to the available reliability criteria for paleomagnetic measurements (e.g., Van der Voo, 1990; Pechersky and Didenko, 1995), results are considered valid if the accuracy of estimating the age of rocks and the characteristic component of their *NRM* does not exceed 20–30 Ma. In our set of specimens, only 17 independent measurements were found to comply with the required accuracy in determining the age (Fig. 9b; Table 3). One of them on the Kuonamka Dykes of the Anabar shield (Ernst et al., 2000), despite its

Table 3
Selected 1900–1500 Ma Siberian paleomagnetic poles.

No.	Description	Age	+/−	Dec	Inc	a_{95}	Glat	Glong	Plat	Plong	d_p	d_m	Author
*1	Basites, Olenek uplift	1473	24	28.8	32.6	12.2	71.5	106.5	33.6	253.1	7.6	14.3	Wingate et al. (2009)
2A	Gonam Fm, Uchur–Maya area	1485	35	31.5	10.0	8.7	57.0	131.3	32.1	273.6	4.3	8.5	Veselovsky et al. (2009)
$2A_r$	Gonam Fm, Uchur–Maya area								33.8	303.3			
2B	Gonam and Kunkuli Fms, Uchur-Maya area	1506	109	150.0	15.0	16.0	57.5	135.0	−20.4	166.9	8.4	16.4	Pisarevsky (2005) (#5662)
$2B_r$	Gonam and Kunkuli Fms, Uchur-Maya area								−10.1	190.6			
3A	Burdur Fm, Anabar massif	1500	150	17.0	−28.9	3.2	71.3	107.0	2.5	270.6	1.9	3.5	Veselovsky et al. (2009)
3B	Upper part of the Burdur Fm, Anabar massif	1595	95	39.0	43.7	5.7	71.7	108.0	39.0	241.0	4.5	7.0	Gurevich (1983)
4A	Kotuikan Fm, Anabar massif	1500	150	12.9	−36.3	7.9	71.7	108.0	−2.3	275.9	5.4	9.2	Veselovsky et al. (2009)
4B	Kotuikan Fm, Anabar massif	1470	50	178.0	23.0	10.0	70.6	106.0	−7.0	108.0	4.0	7.0	Pisarevsky (2005) (#5591)
5	Kuonam dykes, Anabar massif	1503	5	56.8	−9.5	28.0	70.0	110.0	6.0	234.0	14.0	28.0	Ernst et al. (2000)
6	Intrusive North slope, Anabar massif	1513	51	27.9	5.3	9.9	71.5	106.5	18.9	256.9	5.0	9.9	Veselovskiy et al. (2006)
7	Sololi sill, Olenek uplift	1530	270	67.0	30.0	4.0	71.0	124.0	−23.0	51.0	2.5	4.4	Pisarevsky (2005) (#5657)
8	Intrusions, Aldan shield	1590	100	225.0	38.0	13.0	58.0	126.0	−3.0	84.0	9.1	15.4	Pisarevsky (2005) (#5509)
8_r	Intrusions, Aldan shield								−7.3	107.1			
9	Il'inskaya Fm., Anabar massif	1595	95	353.0	−19.7	2.9	71.7	108.0	8.0	295.0	1.6	3.0	Gurevich (1983)
*10	Dykes of Chaya Complex, Baikal ridge	1674	29	6.6	17.4	5.5	55.0	109.0	43.6	280.0	3.2	5.3	Vodovozov et al. (2007)
*11	Dykes of Chaya Complex, Akitkan ridge	1674	3	193.3	−16.4	5.1	57.6	110.8	39.6	272.0	2.7	5.3	Vodovozov et al. (2007) Vodovozov (2010)
*12	Dykes of Chaya Complex, Akitkan ridge	1752	293	189.4	−16.3	8.6	56.1	109.3	41.6	276.0	6.2	12.8	Vodovozov et al. (2007) Vodovozov (2010)
13	Granites, Ulkan massif	1719	10	46.8	39.8	4.7	56.3	134.5	42.1	249.4	3.4	5.6	This study
$*13_r$	Granites, Ulkan massif								48.2	281.6			
14	Elgetey Fm., Ulkan massif	1732	4	122.4	46.4	12.8	56.3	134.6	7.1	183.5	10.5	16.4	This work
14_r	Elgetey Fm., Ulkan massif								18.4	205.0			
*15	Granites, Angar–Kan inlier	1734	5	348.4	19.5	7.0	56.8	94.0	42.9	289.0	3.8	7.3	Didenko et al. (2009a)
16	Anorthosites (A-comp.), Dzhugdzhur massif	1736	6	112.0	13.0	13.4	55.7	134.0	−7.0	202.0	7.0	14.0	Mikhailova et al. (1994)
16_r	Anorthosites (A-comp.), Dzhugdzhur massif								4.6	224.1			
17	Metamorphites, Aldan shield	1900		35.7	−29.3	8.5	57.6	131.9	11.0	277.0	5.2	9.4	Pavlov (2006)
17_r	Metamorphites, Aldan shield								12.6	301.6			
18	Metamorphites, Anabar massif	1900		9.8	−11.5	6.9	70.0	108.0	13.9	278.0	3.6	7.0	Savrasov (1990)
19	Metamorphites, Anabar massif	1900		168.0	4.0	13.0	70.0	111.0	−16.0	125.0	6.5	13.0	Pisarevsky (2005) (#5511), Age from Pavlov (2006)
*20	Shumikhin granite (Toisuk), Sharyzhalgai inlier	1844	15	163.9	26.0	8.0	52.2	103.2	22.6	300.0	4.7	8.6	Didenko et al. (2006)
*21	Khibelen Fm. (Zavorotnaya), Baikal ridge	1849	11	326.3	−0.7	6.5	55.0	108.0	28.2	327.0	3.3	6.5	Vodovozov et al. (2007)
*22	Okun' Fm, Akitkan ridge	1850		358.6	−8.1	9.6	57.6	110.8	28.5	291.0	4.9	9.7	Vodovozov (2010)
*23	Shumikhin granite (Orlyonok), Sharyzhalgai inlier	1855	5	9.2	−26.2	5.7	52.0	104.1	23.7	274.0	3.3	6.2	Didenko et al. (2006)
24	Chaya Fm., Akitkan ridge	1863	9	192.5	18.6	6.9	57.6	110.8	22.1	277.5	3.7	7.2	Didenko et al. (2009b) Vodovozov (2010)

Table 3 (*Continued*)

No.	Description	Age	+/−	Dec	Inc	a_{95}	Glat	Glong	Plat	Plong	d_p	d_m	Author
*24$_t$	Chaya Fm., Akitkan ridge	1854							22.1	277.5			
25	Khibelen Fm, Akitkan ridge	1855	4	207.4	16.1	7.3	57.6	110.8	20.5	261.0	3.9	7.5	Didenko et al. (2009b) Vodovozov (2010)
*25$_t$	Khibelen Fm, Akitkan ridge	1859							20.5	261.0			
26	Malaya Kosa Fm., Baikal ridge	1878	4	188.7	8.1	5.0	54.7	108.8	−30.8	98.7	2.5	5.0	Didenko et al. (2009b)
27	Khibelen Fm., Baikal ridge	1878	4	341.2	−28.5	4.3	55.0	108.0	18.0	307.1	2.6	4.7	Vodovozov et al. (2007)
28	Intrusions, Olenek uplift	1965	115	26.0	28.0	7.0	71.0	124.5	32.0	274.0	4.2	7.7	Pisarevsky (2005) (#5524)

Glat, Glong, Plat, Plong are the latitude and longitude sampling area and paleomagnetic pole position, respectively (in°); d_p and d_m are the semi-axes of the 95% confidence ellipse about the pole position (in°). The bottom index "r" means the pole position after applying the correction for rotation of the Aldan–Stanovoy province relative to the Angar–Anabar one in the Middle Paleozoic (coordinates of the Euler pole: 62° N, 117° E, angle of rotation = + 25°) according to Pavlov et al. (2008); 2 – the bottom index "t" denotes the age accepted for calculating the APWP (see the text in more detail); the top index "*" means paleomagnetic poles used for calculating the APWP for Siberia in the late Paleoproterozoic–early Mesoproterozoic (Fig. 11a; Table 4). For the data from GPDB-4.6 (Pisarevsky, 2005) numbers (RESULTNO) are specified. Other designations see in Table 1.

satisfactory accuracy by age – ±5 Ma, has poor paleomagnetic accuracy (a_{95} = 28°), whereas according to Van der Voo (1990) this value should be no more than 16°.

Of the remaining 16 measurements, two on rocks of the Baikal ridge with an age of 1878 ± 4 Ma were excluded (Table 3, entries 25 and 26). The paleomagnetic directions of the two similar-age rock formations and hence the positions of their paleomagnetic poles calculated based on these directions differ significantly,

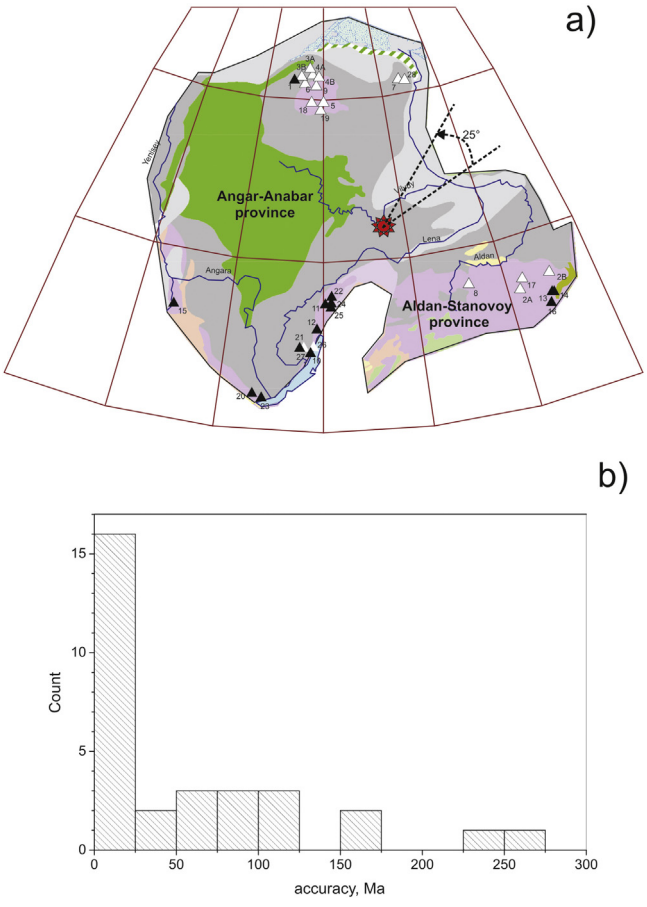


Fig. 9. Distribution of Paleoproterozoic and early Mesoproterozoic paleomagnetic data on the Siberian platform (a) and accuracy of defining the age of the NRM characteristic component (b). Solid (open) symbols mean reliable (unreliable) paleomagnetic data (see text). Numbers on (a) correspond to numbers in Table 3.

which is probably due to their formation outside a single tectonically rigid block; the difference of paleolatitudes is 11.1 ± 3.7° and the magnitude of the relative rotation is 27.5 ± 3.8°. This agrees with the conclusion made by Gladkochub et al. (2006): "Based on the findings obtained as a result of studying indicator mafic-ultramafic and granitoid complexes within the southern part of the Siberian craton, it became possible to conclude that by ~1.85 Ga the formation of the southern Siberian craton structure had completed".

Thus, according to these two criteria only 14 measurements remained (solid symbols in Figs. 9a and 10) to be used for further analysis and generation of the APWP for Siberia at the end of the Paleoproterozoic–beginning of the Mesoproterozoic. Most of the data relates to the Angar–Anabar province (Akitkan and North Baikal ridges, Sharyzhalgai and Angar–Kan salients of the Siberian platform basement, Anabar massif) and only three measurements to the Aldan–Stanovoy province (Fig. 9a), two of them being the result of the present study (Table 3, entries 13 and 14) and one belonging to Mikhailova et al. (1994) which is on anorthosites of the Dzhugdzhur massif (Table 3, entry 16).

Two corrections were introduced for the purposes of the further analysis. First, new positions of paleomagnetic poles of rock formations from the Aldan–Stanovoy province by rotating them by 25° counterclockwise about the rotation pole with the coordinates: 62° N, 117° E having regard to the rotation of the Aldan–Stanovoy block relative to Angar–Anabar during the opening of the Vilyui rift in the middle Paleozoic (Pavlov et al., 2008). In Table 3 and Fig. 10 they are marked with index "r". Second, a minor age correction was applied to the Khibelen (Table 3, entry 25) and Chaya (Table 3, entry 24) formations of the Akitkan ridge. The absolute geochronological age (U–Pb, zircon) of the former is estimated at 1855 ± 4 Ma, and that of the latter – 1863 ± 9 Ma (Didenko et al., 2009a,b; Vodovozov, 2010). In fact the Khibelen Formation is stratigraphically lower than the Chaya Formation therefore to generate APWPs for them their age was defined as 1859 and 1854 Ma, respectively, that is, in the first case the maximum possible within the accuracy of determination and in the second case – the minimum possible within the accuracy of determination. In Table 3 and Fig. 10 they are marked with index "t".

We believe it is necessary to specifically comment on the position of the paleomagnetic poles obtained for the Ulkan massif in the present study and anorthosites of the Dzhugdzhur massif (Mikhailova et al., 1994).

Coordinates of the original pole of the Ulkan granitoid complex (Plat = −42.1°, Plong = 69.4°, d_p = 3.4°, d_m = 5.6°) do not coincide with the coordinates of the pole of the similar-age

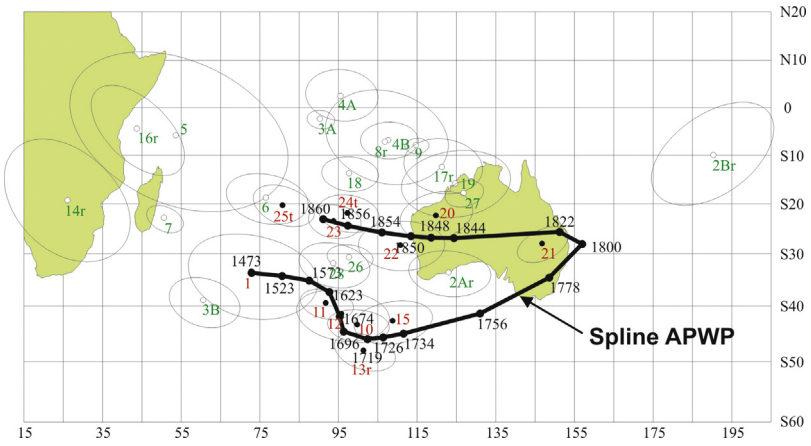


Fig. 10. Distribution of Paleoproterozoic and early Mezoproterozoic paleomagnetic poles of the Siberian platform. Solid (open) symbols mean reliable (unreliable) paleomagnetic poles (see text). Numbers on (a) correspond to numbers in Table 3. The APWP was generated by the cubic spline method (spline-factor is 5000) under the GMAP programme (Torsvik and Smethurst, 1999). For the spline APWP coordinates see Table 4.

Angar–Kan granitoid block but after applying the correction for the rotation of the Aldan–Stanovoy block relative to the Angar–Anabar one in the middle Paleozoic (Pavlov et al., 2008) their positions become very close (entries 13_r and 15, Table 3; Fig. 10). Firstly, this coincidence, in our opinion, is evidence in favour of the Paleoproterozoic age of the HTC of granites from the Ulkan complex. Secondly, it suggests that the Ulkan block at the time of acquiring the characteristic HTC of the *NRM* of granites from the Ulkan complex was rigidly connected to the Angar–Anabar province. Only in the Middle Paleozoic did they experience a significant rotation relative to each other during the opening of the Vilyui basin (aulacogen, rift) (Pavlov et al., 2008). The pole of the Ulkan granites is used to generate the APWP for Siberia.

We see a different situation for paleomagnetic poles of volcanogenic-sedimentary rocks of the Elgetey Formation (this study) and anorthosites of the Dzhugdzhur massif (Mikhailova et al., 1994). Their coordinates both before the rotation about the above-mentioned pole of rotation (Table 3) and after the rotation are in South Africa and far away from the similar-age granitoids of the Angar–Kan block (entries 14_r and 16_r, Table 3; Fig. 10). Given the paleomagnetic reliability of the poles under consideration, we believe one can speak about tectonic incoherence of the Ulkan and Dzhugdzhur massifs at ~1732 Ma relative to the main body of the Siberian craton, that is, the Angar–Anabar province. Poles of volcanogenic-sedimentary rocks of the Elgetey Formation (this study) and anorthosites of the Dzhugdzhur massif (Mikhailova et al., 1994) are therefore not used in generating the APWP for Siberia.

Positions of all the paleomagnetic poles from Table 3 are shown in Fig. 10 after reversing them to one polarity according to the principle of minimizing the movement. To the early Proterozoic poles we added the pole position defined by Wingate et al. (2009), corresponding to the beginning of the Mesoproterozoic (Table 3), with the result that we get, in our opinion, a set of 12 most reliable paleomagnetic poles to generate the late Paleoproterozoic–early Mesoproterozoic segment of the APWP for Siberia. Analysis of their distribution yields two clusters with reliably dated poles (red numbers and solid symbols in Fig. 10): (1) 1840–1860 Ma and (2) 1740–1470 Ma; within these clusters one can observe a regular decrease in age of the poles directed in the branches of the loop towards each other. Based on these 12 poles, the late Paleoproterozoic–early Mesoproterozoic segment of the APWP for Siberia was computer-calculated using the GMAP software (Torsvik and Smethurst, 1999), which is shown by the solid line in Fig. 10, and coordinates are listed in Table 4.

Table 4
Apparent polar wander path for Siberia in the Late Paleoproterozoic–Early Mesoproterozoic.

Plat,°	Plong,°	Age, Ma
−33.6	73.1	1473
−34.2	80.9	1523
−35.1	87.8	1574
−36.2	90.3	1599
−37.3	92.9	1624
−41.8	95.5	1674
−44.5	96.5	1697
−45.9	102.6	1719
−45.6	106.6	1727
−44.9	111.7	1734
−41.3	131.2	1756
−34.5	148.8	1778
−27.9	157.1	1800
−25.6	151.4	1822
−26.8	124.6	1844
−26.8	122.7	1845
−26.7	120.8	1846
−26.7	118.8	1847
−26.6	116.7	1848
−26.4	113.9	1849
−26.3	112.7	1850
−26.0	109.4	1851
−25.7	106.3	1853
−25.2	103.1	1854
−24.9	100.8	1855
−24.3	97.6	1856
−23.7	94.5	1858
−23.0	91.3	1860

Poles positions were calculated by the cubic spline method (smoothing spline is 5000) under the GMAP programme (Torsvik and Smethurst, 1999). Other designations see in Table 3.

5.2. Polarity and reconstructions of Siberia and Laurentia' mutual position

In contrast to Laurentia poles, to define polarity (normal or reversal) for the Paleoproterozoic poles of Siberia is a challenge as we lack a high-quality set of poles older than the late Precambrian. There are gaps in one or more million years both for Neoproterozoic and Mesoproterozoic times. Moreover, there are two alternative models for the Neoproterozoic segment of the APWP for Siberia. The first model was proposed by Smethurst et al. (1998), according to which the Neoproterozoic normal polarity poles for Siberia occur in temperate and tropical latitudes of the South Pacific (Fig. 11). Later Shatsillo et al. (2005, 2006) reported new paleomagnetic data on Vendian-Lower Cambrian sedimentary rocks of the southern

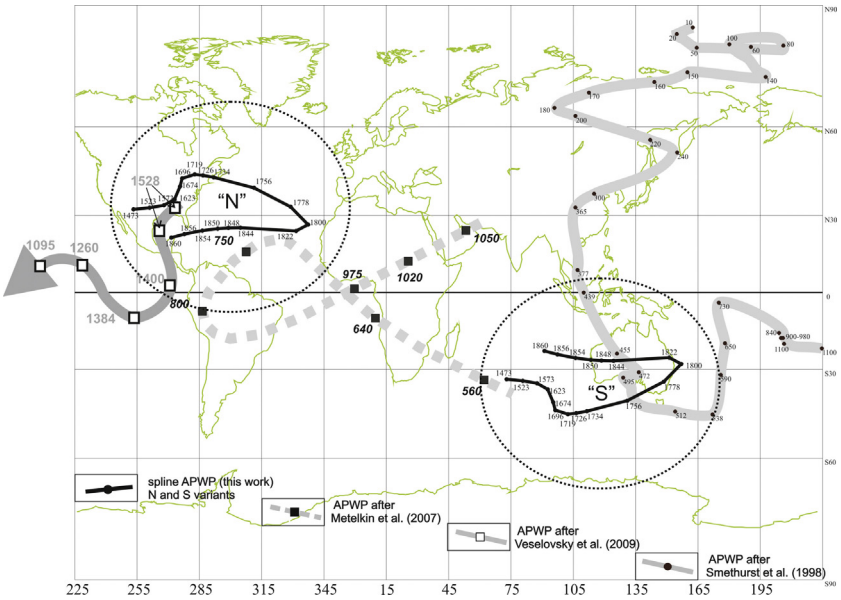


Fig. 11. Comparison of the existing APWP with the one generated as a result of the study. Signs "N" and "S" for the spline APWP (this study) denote two possible variants according to the accepted polarity (see text).

margin of Siberia and showed that corresponding paleomagnetic poles of Siberia of this age are located in high, near Antarctica, and temperate latitudes of the Indian Ocean. With this in mind it was concluded: "The *results* of this study proved that the Vendian segment of the APWP for Siberia resided in the Indian Ocean and that the earlier views concerning the Pacific trends of the Late Riphean and Vendian poles were in error" (Shatsillo et al., 2006).

The second, alternative model of the Neoproterozoic segment of the APWP for Siberia (Fig. 11) was proposed by Metelkin et al. (2007). To construct the model, both original and published data were used including the data obtained by Shatsillo et al. (2005, 2006). The position of the Neoproterozoic poles of normal polarity according to Metelkin et al. (2007) is essentially different from that defined by Smethurst et al. (1998). The second distinctive feature of the trajectory drawn by Metelkin et al. (2007) is the loop which both in age and shape is similar to the Grenvillian loop of the APWP for Laurentia (Pisarevsky and Natapov, 2003).

As mentioned above, the model of the Mesoproterozoic segment of the APWP for Siberia has been proposed by Veselovsky et al. (2009). According to this model, the corresponding normal polarity poles are arranged in the subequatorial zone of the eastern part of the Pacific (Fig. 11) and are, in fact, a continuation of the Neoproterozoic track of the APWP (Smethurst et al., 1998).

An extended discussion of "pros and cons" of the above models of the APWP for Siberia is beyond the scope of this paper. The main purpose of presenting these three models in the paper is to highlight the fact that, as of now, there are no direct paleomagnetic data on polarity of the currently available paleomagnetic poles for the second half of the Neoproterozoic, Mesoproterozoic and, of course, Paleoproterozoic. To choose the option of polarity of Siberia Paleoproterozoic and Mesoproterozoic poles, at this point, we believe, non-paleomagnetic "tools", such as geological, structural, geochronological, and others, should be used as it is shown in a number of studies (e.g., Didenko et al., 2005; Pavlov et al., 2002; Pisarevsky and Natapov, 2003) of which, in our case, a hypothesis of the late Paleoproterozoic supercontinent, can be an example (e.g., Condie, 2002; Hoffman, 1997; Meert, 2002, 2012; Pesonen et al., 2003; Piper, 2000; Zhao et al., 2002, 2004, 2011).

Possible positions of continental blocks in the structure of the Proterozoic supercontinent based on the paleomagnetic data are subject to ongoing discussions (e.g., Evans and Mitchell, 2011; Pesonen et al., 2003; Pisarevsky et al., 2014; Zhang et al., 2012). Siberia, whose possible positions relative to Laurentia at four time periods are shown in Fig. 12, is no exception. Sign "N" denotes an option when poles taken as northern are located in the North Atlantic and North America, and sign "S" – in the Indian Ocean and Australia (Fig. 11). Original poles were used in calculating positions of Laurentia and Siberia (Fig. 12) with the exception of Siberia at 1590 Ma.

The following paleomagnetic poles were used to position Laurentia (Fig. 12) at (1) 1840–1860 Ma from the A comp. of the Wathaman batholith (Plat = 9, Plong = 293; Symons, 1991); (2) 1730–1750 Ma from the A component of the Molson dykes (Plat = −16, Plong = 277; Halls and Heaman, 2000) and Cleaver dykes (Plat = 19, Plong = 277; Irving et al., 2004); and (3) 1590 Ma from dykes of Western Channel Diabase (WCD) (Plat = 9, Plong = 245; Irving et al., 2004; age by Hamilton and Buchan, 2010); and (4) 1470–1480 Ma from the St. Francois Mtn. (Plat = −13, Plong = 219; Meert and Stuckey, 2002). Usually normal polarity is attributed to these poles, that is, they are northern, which is consistent with the APWP for Laurentia constructed more than 20 years ago (Piper, 1982), although there are alternative opinions (Park, 1994). Our constructions were also based on normal polarity of the above mentioned poles of Laurentia, as the reverse polarity option requires a more detailed analysis and revision of the existing set of the Precambrian poles, which is not the subject of this paper.

The 1860–1840 Ma reconstructions place Laurentia in the northern hemisphere, its modern northern edge being at 15–18° and facing south (Fig. 12a). The position of Siberia calculated from the coordinates of the paleomagnetic pole located in the Atlantic Ocean is somewhat lower in latitude; it takes the equatorial position of 5° N to 15° S and its modern northern margin faces north towards Laurentia ("N" on Fig. 12a). The paleocontinents are within 7–8°. In the case of the other polarity option when the north pole is in the Indian Ocean, the position of Siberia will change by 180°, that is, the southern (Baikalian) margin of the Siberian craton will be by the Arctic margin of Laurentia at 8–10° N, and the two cratons will be at a minimum distance ("S" on Fig. 12a).

We have the following position of the blocks at 1730–1750 Ma (Fig. 12b) according to the pole from Halls and Heaman (2000).

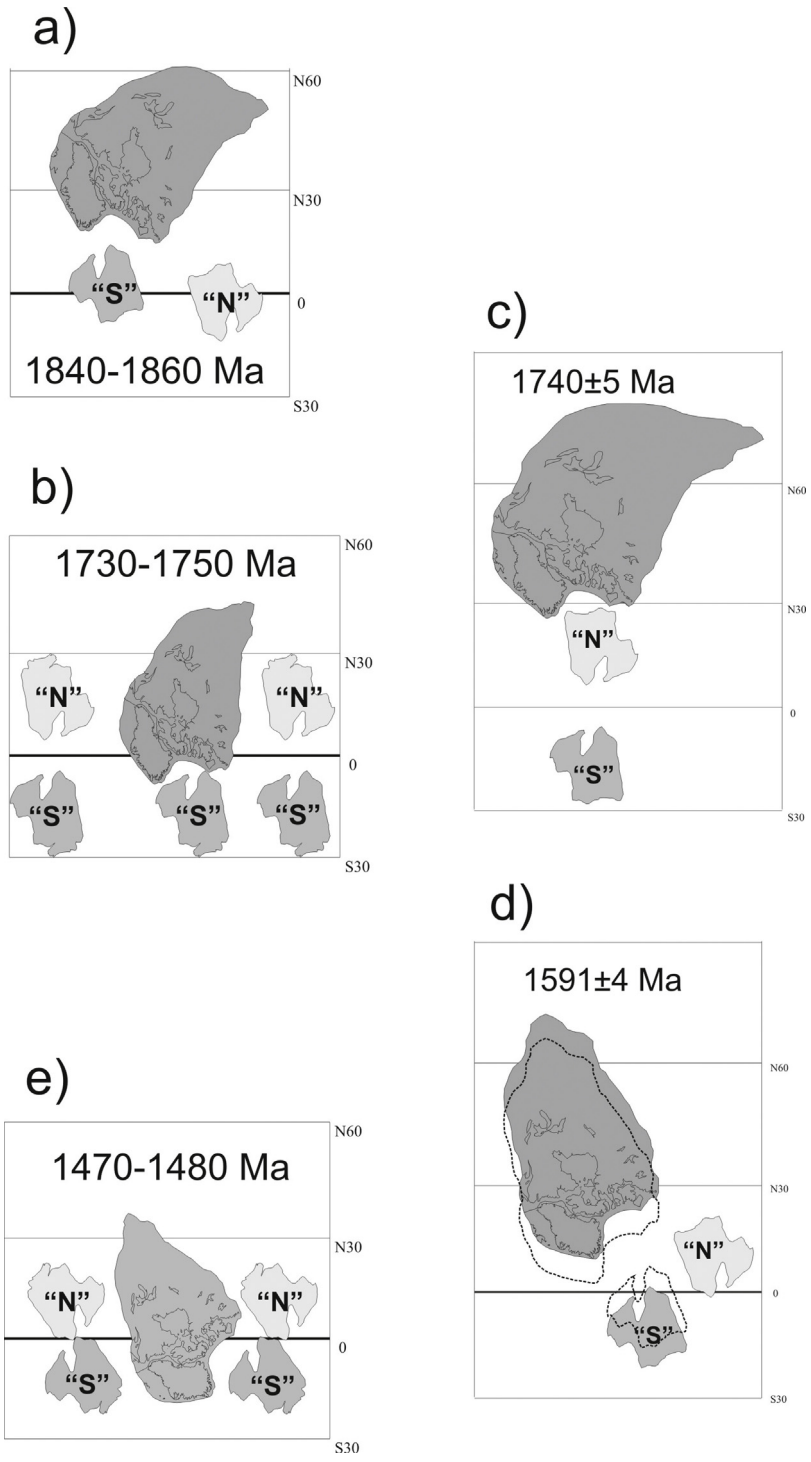


Fig. 12. Possible positions of Siberia relative to Laurentia according to the existing paleomagnetic poles 1840–1860 Ma (a), 1730–1750 Ma (b), 1740 ± 5 Ma (c), 1591 ± 4 Ma (d) and 1470–1480 Ma (e). In (d) dashed line shows the position of Siberia and Laurentia considering the A95 confidence angle. Paleomagnetic poles used to calculate positions of Laurentia and Siberia: (a) A comp. Wathaman batholiths – Plat = 9, Plong = 293 (Symons, 1991) and Shumikhinsky granites – Plat = −23.7, Plong = 94 (No. 23, Table 3); (b) A comp. Molson dykes – Plat = −16, Plong = 277 (Halls and Heaman, 2000) and Angar–Kan granites – Plat = −42.9, Plong = 109 (No. 15, Table 3); (c) Cleaver dykes – Plat = 19, Plong = 277 (Irving et al., 2004) and Angar–Kan granites – Plat = −42.9, Plong = 109 (No. 15, Table 3); (d) from dykes of Western Channel Diabase – Plat = 9, Plong = 245 (Irving et al., 2004; age by Hamilton and Buchan, 2010) and the interpolated pole for Siberia (Table 4); (c) St. Francois Mtn., Plat = −13, Plong = 219 (Meert and Stuckey, 2002) and Sololi intrusions (Wingate et al., 2009) – Plat = −33.6, Plong = 73.1 (No. 1, Table 3).

Laurentia moved by about 20° southward and simultaneously rotated by 30–35° counterclockwise. In the case of the north pole location in the Atlantic Ocean Siberia will be entirely in the subequatorial latitudes of the northern hemisphere from 5 to 30°. The orientation of Siberia remains virtually unchanged compared to its position at 1840–1860 Ma. In this case, joining of Siberia and Laurentia could occur along their current western margins as well as modern eastern ones ("N" on Fig. 12b). In the case of the other polarity option (north pole in the Indian Ocean) the position of Siberia will change by 180°, that is, the southern (Baikalian) edge

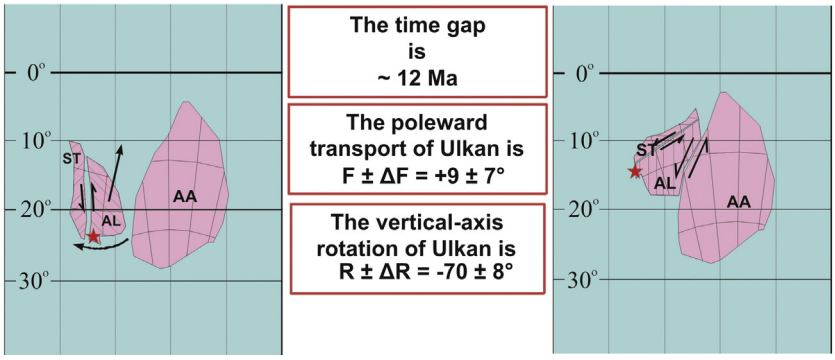


Fig. 13. Reconstructions of Stanovoy (ST), Aldan (AL), and Angar–Anabar (AA) provinces of Siberia at 1732 Ma (left) and 1720 Ma (right). Star denotes the study area.

of Siberia will be by the modern Arctic margin of Laurentia at 5–8° S and the Arctic one at 30° S ("S" on Fig. 12b). In this case, the present-day Arctic margin of Laurentia and the southern Baikalian margin of Siberia could become adjacent, that is, their position is not in conflict with the hypothesis of them being part of the supercontinent.

According to the Cleaver dykes pole (Irving et al., 2004), for the same time period (1740 Ma) the position of Laurentia has not changed much during the past 100 million years – it drifted northward within 10° and rotated insignificantly counterclockwise. In accordance with this pole, the present-day Arctic margin of Laurentia was at 30° N (Fig. 12c). The position of Siberia at this time according to our data is described above. In such a situation, these cratons could become adjacent along their present-day Arctic margins only provided that the north pole (normal polarity) for Siberia is in the Atlantic (Fig. 12c, option "N"). In the case that the North Pole is in the Indian Ocean, there had to be a gap as wide as 30° (Fig. 12c, option "S").

The WCD pole (Irving et al., 2004) and age interpretation after (Hamilton and Buchan, 2010; Buchan, 2014) for Laurentia and the interpolated pole for Siberia (Table 4) due to the lack of reliable data for this interval (Table 3), were used to reconstruct the mutual position of the two cratons at 1590 Ma (Fig. 12d). At this time, Laurentia was entirely in the northern hemisphere, its Arctic margin in the tropical latitudes of 10–25° facing the south-east. Siberia, in the case of the north pole being in the Atlantic Ocean, will be entirely in the subequatorial latitudes of the northern hemisphere from 0 to 20°, its Arctic margin facing the north-west towards the Arctic margin of Laurentia (Fig. 12d, option "N"). In the case of the other polarity option (north pole in the Indian Ocean), the position of Siberia will change by 180° and will be in the subequatorial latitudes of the southern hemisphere. The "Baikalian" margin of Siberia will face the north-west towards the Arctic margin of Laurentia (Fig. 12d, option "S"). In the latter case there is a latitudinal gap of about 15° between the Arctic margin of Laurentia and the "Baikalian" margin of Siberia. But considering A95 = 6° (confidence limit) for the Laurentia original pole; Irving et al., 2004) and similar A95 values for the Siberia interpolated pole one can significantly bring together these cratons and "close" the gap (Fig. 12d, dashed line).

We have the following position of the blocks at 1470–1480 Ma (Fig. 12e). Laurentia moved by about 10° further south rotating at the same time by approximately 20° counterclockwise. Siberia in the case of the north pole being in the Atlantic Ocean will be entirely in the subequatorial latitudes of the northern hemisphere from 0 to 20° (Fig. 12e, option "N"). In the case of the other polarity option (northern pole in the Indian Ocean), the position of Siberia will change by 180°, and again the southern (Baikalian) margin of Siberia will be contiguous with the present-day Arctic margin of Laurentia (Fig. 12e, option "S").

If we proceed from the assumption that Laurentia, Siberia, Baltica, India and Australia started joining 1.80 Ga ago (Zhang et al., 2012), then one combination – the modern southern edge of Siberia is close to the modern Arctic edge of Laurentia – occurs in three reconstructions at 1840–1860, 1730–1750 and 1470–1480 Ma (Fig. 12a, b, and e). Such fit is not observed in the fourth reconstruction at ca. 1590 Ma, but the difference is not critical. Taking into account the confidence intervals, one can achieve an acceptable position of the cratons when the southern margin of Siberia is contiguous with the Arctic margin of Laurentia (Fig. 12d).

Such a joint presence of Siberia and Laurentia in the structure of the Paleoproterozoic supercontinent is referred to not only in Zhang et al. (2012), where our paleomagnetic data on the Akitkan ridge are used (Didenko et al., 2009a,b), but also in Condie (2002) based on the analysis of geological and isotopic-geochronological data on the end of the late Proterozoic. The same relative position of the two paleocontinents at the turn of the Meso- and Neoproterozoic considering the available geological, structural and isotopic-geochemical data is referred to in Rainbird et al. (1998) and considering the paleomagnetic data in Pavlov et al. (2002) and Metelkin et al. (2012), and on the basis of matching the LIP record between southern Siberia and northern Laurentia (Ernst et al., 2014).

The 1740 Ma reconstruction (Fig. 12c) wherein the position of Laurentia is calculated according to the Cleaver dykes pole (Irving et al., 2004) and the latitudinal gap between the cratons can hardly be "removed" by applying correction for A95 values. We are not able to give a reasonable explanation as the analysis of Paleoproterozoic paleomagnetic poles of Laurentia is beyond the scope of this paper, on the one hand, and on the other, without knowing the detailed geology of Laurentia in the Paleoproterozoic it is unlikely that the authors of this paper are capable of doing it. The only thing worth mentioning here is the presence of at least two (high- and low-latitudinal) groups of paleomagnetic poles in the interval 1850–1740 Ma for Laurentia, which are similar in their meridional position but differ appreciably in their latitudinal position (see Table 1 in Pesonen et al., 2003). The Cleaver dykes pole (Irving et al., 2004) has approximately the same meridional position and the latitudinal position, which is intermediate between the high- and low-latitudinal groups.

On the strength of the data presented here, one can conclude that in the context of the Paleoproterozoic supercontinent model, preference is given to the position of Paleoproterozoic north poles of Siberia in the Indian Ocean and Australia (APWP "S" in Fig. 11), rather than in the Atlantic Ocean (APWP "N" in Fig. 11). This conclusion is valid if we adhere to the accepted polarity option for the

Paleoproterozoic paleomagnetic poles of Laurentia (Piper, 1982) which is also used in (Pesonen et al., 2003; Pisarevsky et al., 2014).

6. Position of the Ulkan massif

Paleomagnetic data on the Paleoproterozoic Ulkan massif made it possible to derive two ancient paleomagnetic directions, close in time of formation to the rocks under study, and to calculate the corresponding paleomagnetic poles for two sections of the late Paleoproterozoic – 1732 Ma for volcanogenic-sedimentary rocks of the Elgetey Formation (positive fold, reversals, and conglomerates tests) and less than 1720 Ma for granites of the Ulkan Complex. According to our data and comparison with the APWP for Siberia (see the previous part), at the time of the studied rocks formation the Ulkan massif was located at 18–26° S. The massif rotated about the vertical axis by 70 ± 8° clockwise 1732 Ma (age of the Elgetey Formation) and ~1720 Ma (age of the HTC of the Ulkan granites) ago (Figure 13). At 1732 Ma the Ulkan massif and other terranes of Angar–Anabar and Aldan–Stanovoy provinces were located in the southern tropical latitude close to one another. During ~12 Myr both blocks shifted northward by an average of 5°. The Angar–Anabar block retained its azimuth orientation, whereas the Ulkan massif (maybe the Aldan–Stanovoy block) experienced significance clockwise rotation and became tectonically coherent to Siberia ~1720 Ma ago.

7. Conclusions

Paleomagnetic studies on Paleoproterozoic rocks of the Ulkan trough yielded two ancient paleomagnetic directions and made it possible to calculate poles corresponding to them for two time intervals in the late Paleoproterozoic: (a) ~1732 Ma for volcanogenic-sedimentary rocks of the Elgetey Formation – Plat = 7.1° (−7.1°), Plong = 183.5° (3.5°), d_p = 10.5, d_m = 16.4° (positive fold test); and (b) less than 1720 Ma for granites of the Ulkan Complex – Plat = 42.1° (−42.1°), Plong = 249.4° (69.4°), d_p = 3.4°, d_m = 5.6°.

The analysis of the existing paleomagnetic data for Siberia in late Paleoproterozoic–early Mesoproterozoic times allowed 12 key poles to be defined (Table 3), based on which the APWP for Siberia was generated (Fig. 11; Table 4).

We can conclude that in the context of the Paleoproterozoic supercontinent model, preference is given to the position of Paleoproterozoic north poles of Siberia in the Indian Ocean and Australia, rather than in the Atlantic Ocean.

According to the new paleomagnetic data and comparison with the APWP for Siberia, at the time of the studied rocks formation the Ulkan massif was located at 18–26° S. The massif rotated about the vertical axis by 70 ± 8° clockwise 1.732 Ga (age of the Elgetey Formation) and ≤1.72 Ga (age of the HTC of the Ulkan granites) ago.

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